

AI-Powered Real-Time Deepfake and Phishing Detection & Defense System

A cross-modal threat verification engine engineered to neutralize AI-orchestrated executive impersonation, voice cloning, and zero-day LLM spear-phishing attacks in real time.

Problem Statement: The Threat Crisis

Perfect Phishing

Attackers are using Large Language Models(LLMs) to write phishing mails that are grammatically perfect and highly convincing, making them indistinguishable from genuine communication.

Executive Deepfakes

Real-time voice cloning and live video likeness synthesis trick employees into authorizing wire transfers or revealing credentials during live executive calls, circumventing multi-factor authentication control.

Latency Gap

Current detection systems simply cannot analyze and flag these AI-generated audio, video, and text manipulations fast enough

Expected Outcome : Target Architecture

Cross-Modal Verification Engine

Ingest and analyze multiple types of data simultaneously – text, audio, and videos – to detect AI manipulation or deepfake artifacts.

Executive Digital Twin

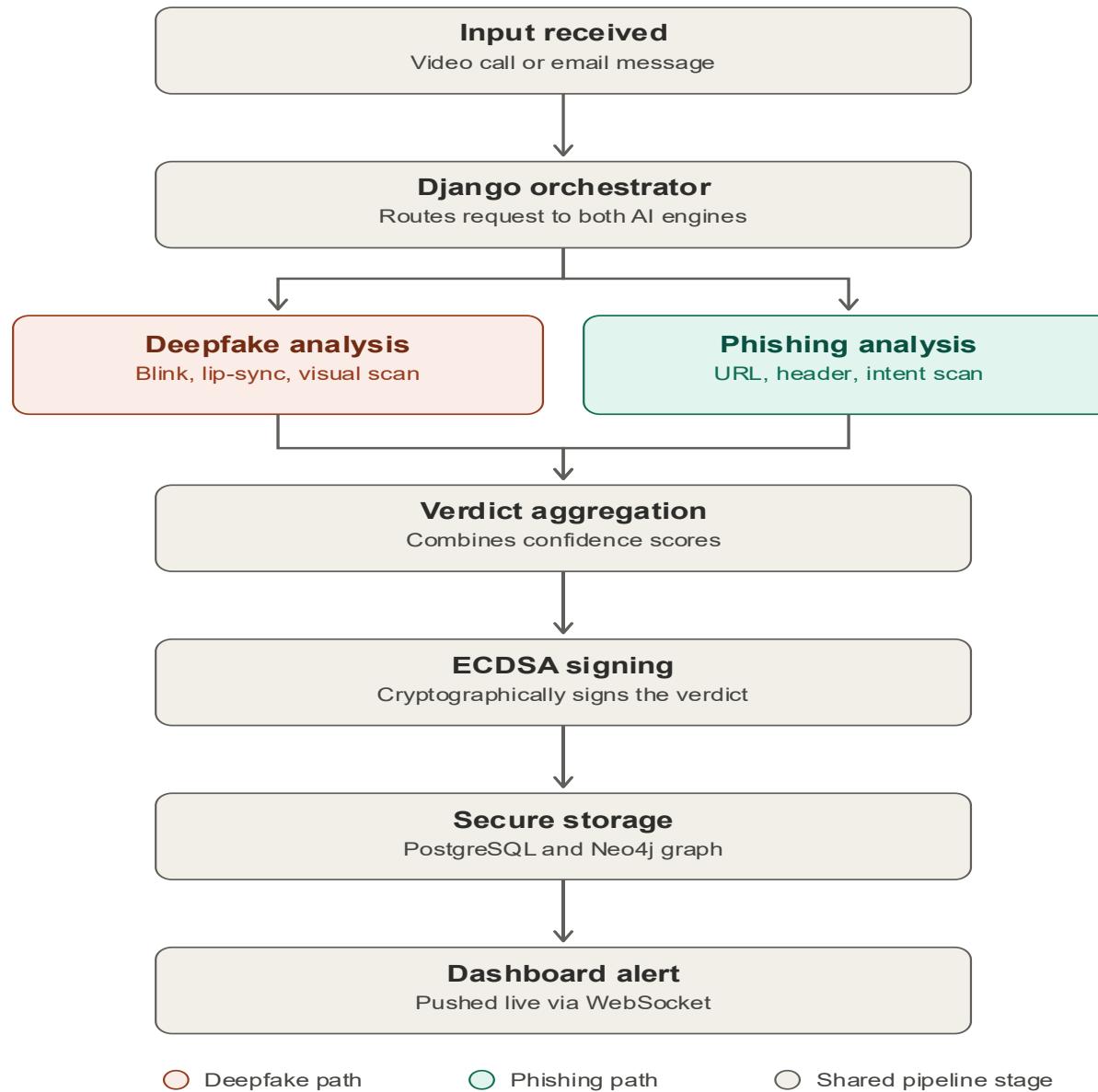
A specialized module designed to recognize baseline communication patterns, voice and visual traits of key executives and instantly flag impersonations.

Neo4j AI Attack Graph

Visualization tool that looks at entire network to connect the dots between seemingly unrelated events.

Security Copilots & Automated Response

Empower security teams with real-time AI copilot telemetry while automatically executing defensive actions when deepfake phishing attempt is verified.



Technologies & Tools

Frontend

Django HTML5 Templates & Vanilla JS

- Native navigator.mediaDevices.getUserMedia for browser-level live stream capture.
- Vanilla JS WebSocket API for real-time video chunk dispatch and alert push.
- D3.js/Chart.js for real-time analytics & graph visualization.

Backend

Django & Django REST Framework

- Django channels (ASGI/Daphne) with Redis datastore for low-latency WebSocket streaming.
- Celery workers for asynchronous AI task dispatch.
- PostgreSQL for relational users, sessions, and audit logs.

AI/ML Engine

MobileNetV2 (PyTorch) & OpenCV

- Librosa for MFCC audio extraction & cross modal temporal lip-sync alignment.
- MediaPipe FaceMesh for blink rate & lip-sync tracking.
- GGUF 4-bit quantized local LLM for phishing intent analysis.

X-factor Layer

ECDSA Cryptography for zero-trust digital signature

- Hardware-backed key signing
- Cryptographic signing of every AI verdict for tamper proof audit trails.
- Sub-millisecond fast-fail rejection for unsigned or spoofed media streams.

Workflow Diagram: System Pipeline

Stage 1 Endpoint Ingestion

- Native browser video/audio capture via navigator.mediaDevices.getUserMedia.
- Client presents ECDSA public key for hybrid authentication.
- Media frame chunks streamed directly via Vanilla JS WebSocket client (ws_client.js).

Stage 2 Gateway & Splitter

- Django Channels validates ECDSA signature.
- Splits media stream into separate audio & frame queues.
- Celery workers dispatch async analysis tasks via Redis channel layer.

Stage 3 Multimodal AI Engine

- Vision: MobileNetV2 detects visuals artifacts in 4-frame batches.
- Audio: Librosa extracts raw audio RMS energy and MFCC features.
- Sync: MediaPipe FaceMesh tracks lip-sync alignment delay and eye aspect ratio (EAR) blink rate.

Stage 4 Threat Correlation

- Combines visual, lip-sync, and blink signals into an aggregated confidence score.
- Stores session data in PostgreSQL and maps attack campaign entities into Neo4j.
- Triggers automated response policies if the score exceeds the threat threshold.

Stage 5 SOC & Mitigation

- Live threat alerts pushed via WebSockets to the server-rendered Django dashboard.
- Renders visual bounding box on artifacts and interactive threat graphs via D3.js
- Automated script immediately terminates the compromised WebSocket session.

The "X-factor": Cryptographic Attestation

Zero-Trust Executive Authentication

While deepfake visuals can be rendered photorealistically, attackers cannot forge executive hardware-backed cryptographic keys without compromising the physical device.

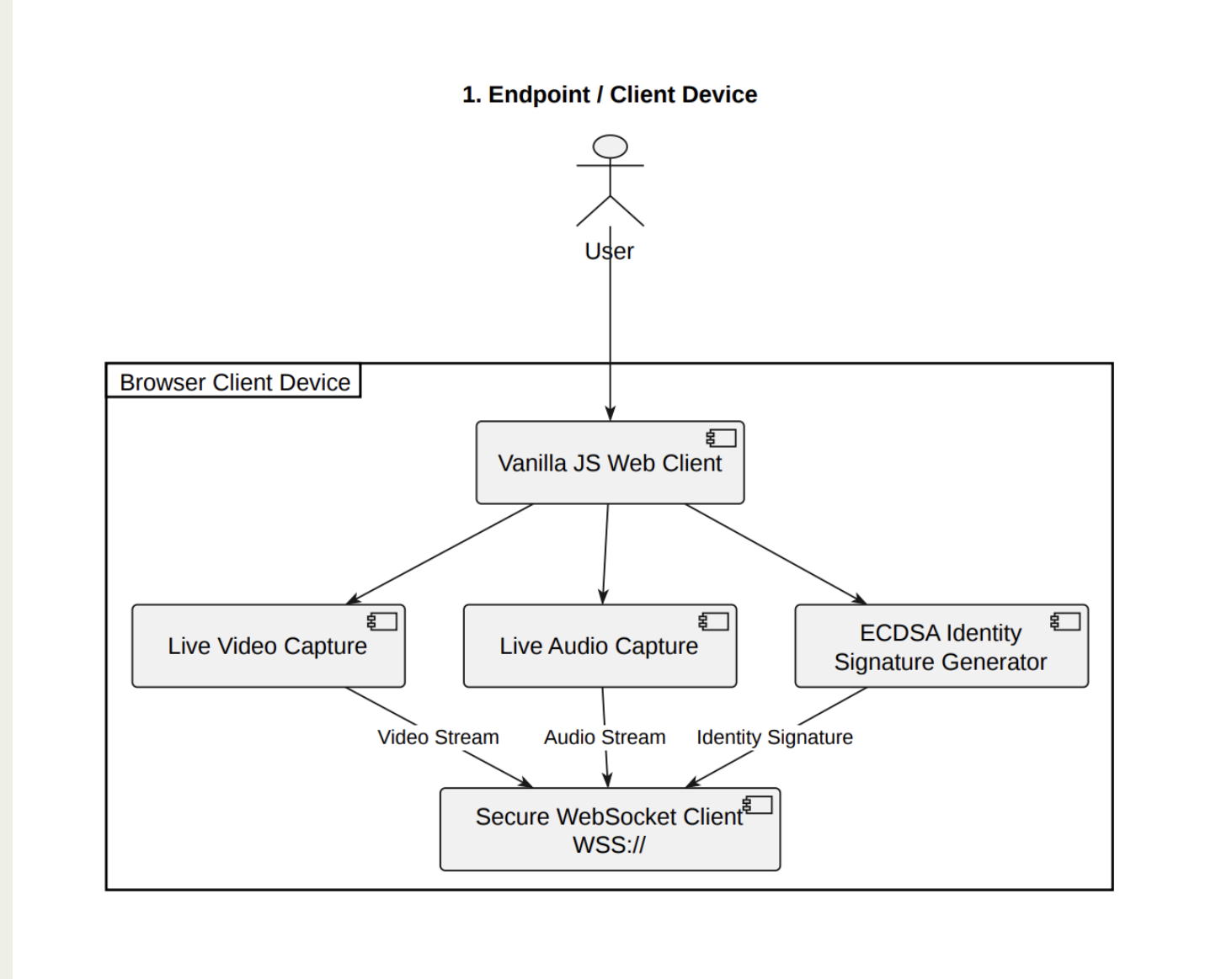
ECDSA Token Generation: Executive devices sign the video stream payload upon connection.

Sub-Millisecond Fast-Fail: Unsigned or spoofed streams fail at the gateway level before expensive GPU inference runs.

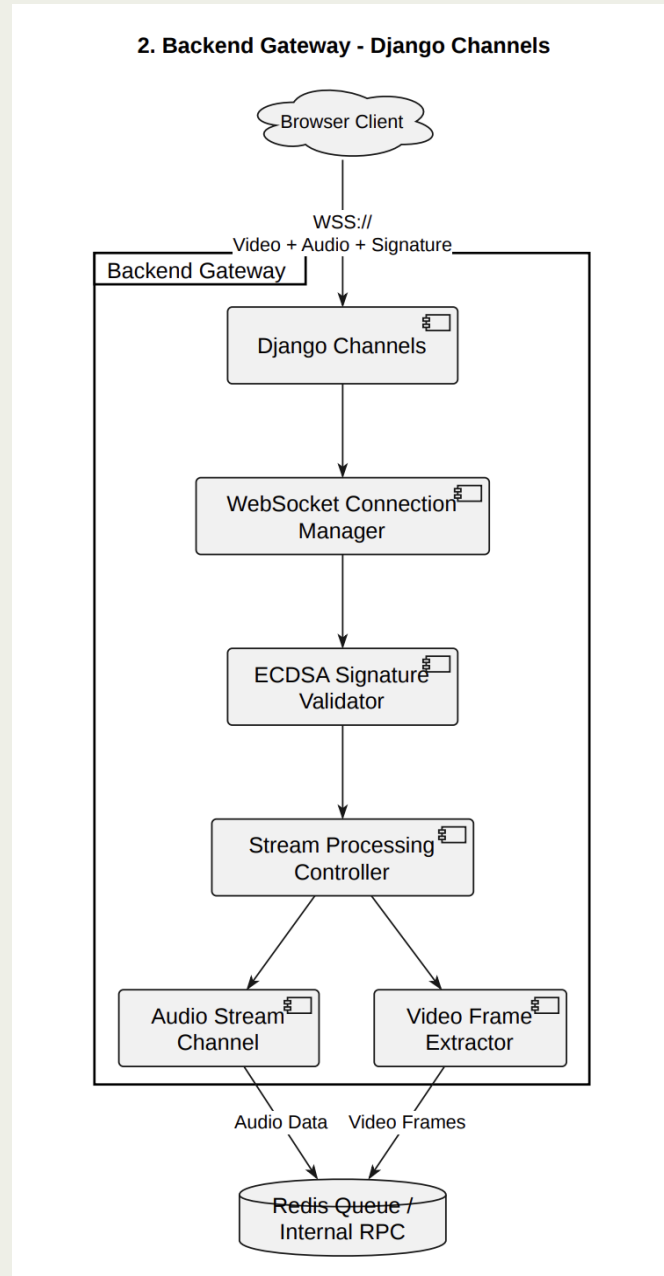
Immutable Identity Baseline: Prevents social engineering over public channels even if the visual rendering is 100% accurate.



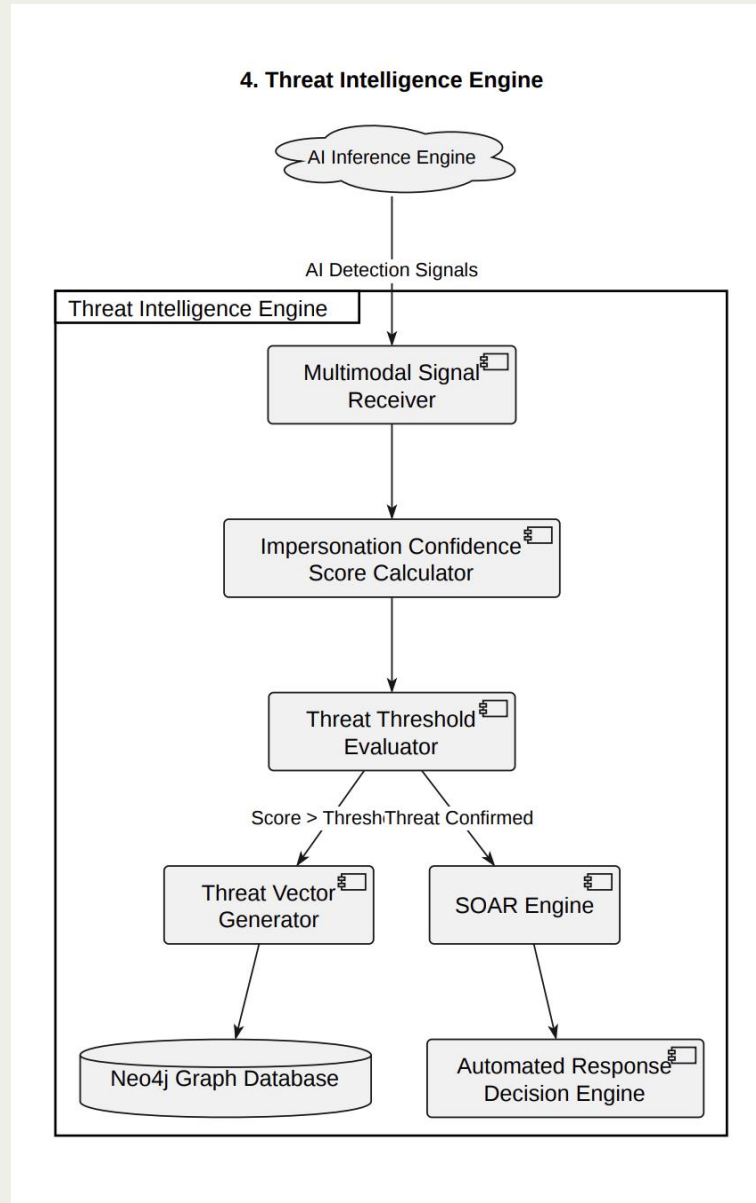
Technical Workflow Diagram:



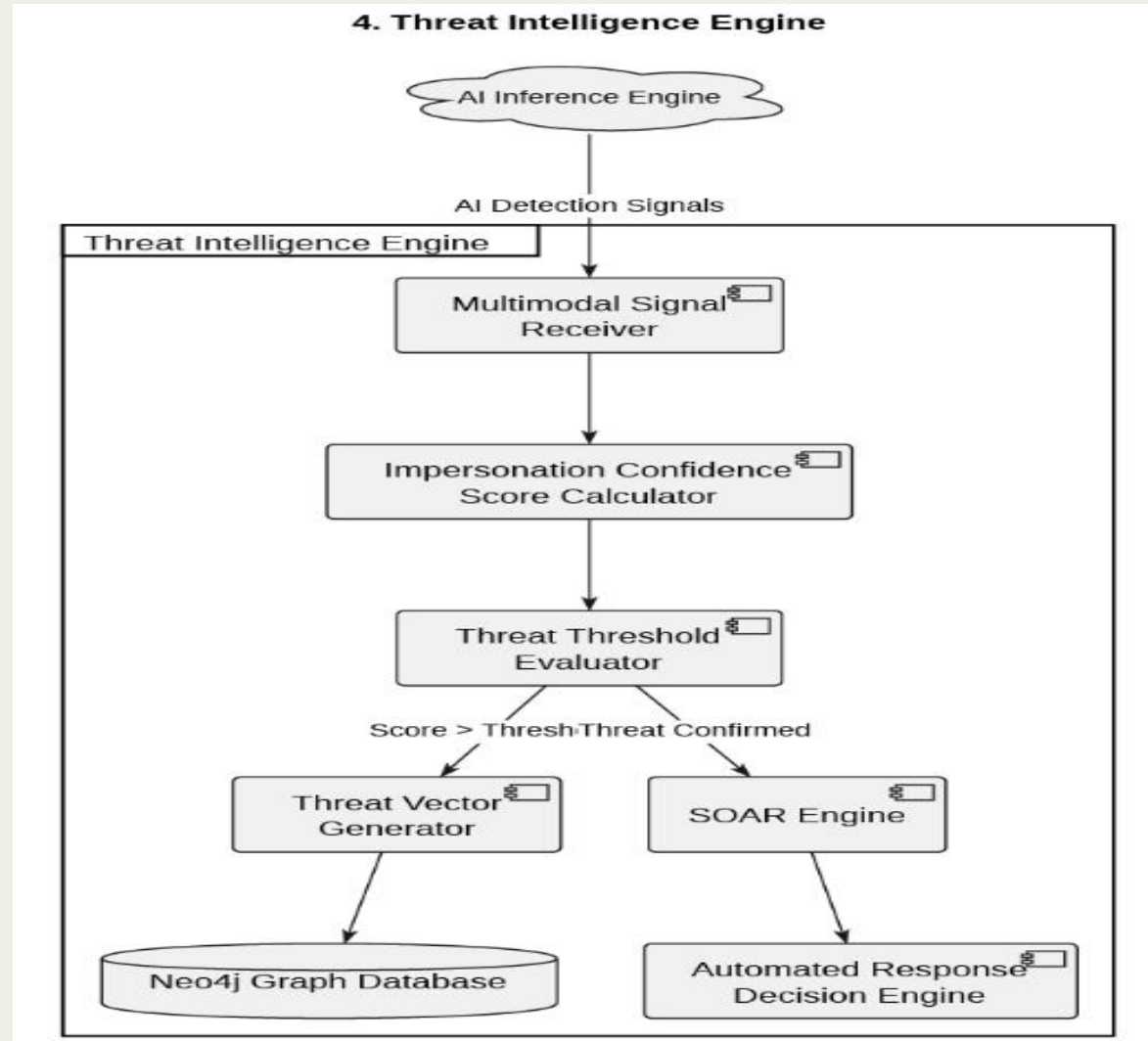
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